KAIZEN MATHS

Lesson starter

Normal Distribution and Standardisation

Standard Normal

Name

S1

For Z ~ N(0, 1), find P(Z < -0.67).

S2

For Z ~ N(0, 1), find P(Z > 0.67).

S3

For Z ~ N(0, 1), find P(-1.10 < Z < 0.67).

S4

For Z ~ N(0, 1), find z such that P(Z < z) = 0.1056.

Show your working

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STA-9E35-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Normal Distribution and Standardisation

Standard Normal

Name

E1

For Z ~ N(0, 1), find P(Z < -1.50).

E2

For Z ~ N(0, 1), find P(Z > 1.75).

E3

For Z ~ N(0, 1), find P(-1.96 < Z < 1.28).

KAIZEN MATHS

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STA-9E35-L1 · E

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STA-9E35-L1 · E

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Standard Normal

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STA-9E35-L1 · E

Show your working

Teacher answers & working

Normal Distribution and Standardisation · Standard Normal

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S1

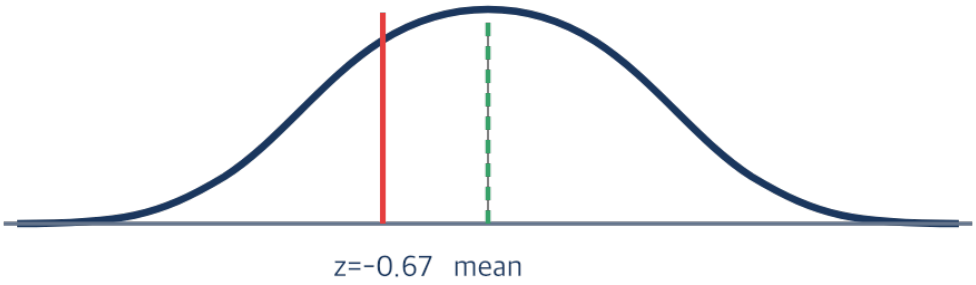
For Z ~ N(0, 1), find P(Z < -0.67).

Answer:

0.2514

Working:

Recognise this as a left-tail standard normal probability.



$P(Z < -0.67) = \Phi(-0.67)$.

Look up $\Phi(-0.67)$ using a standard normal table or calculator.

$\Phi(-0.67) \approx 0.2514$.

0.2514

Teacher answers & working

Normal Distribution and Standardisation · Standard Normal

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S2

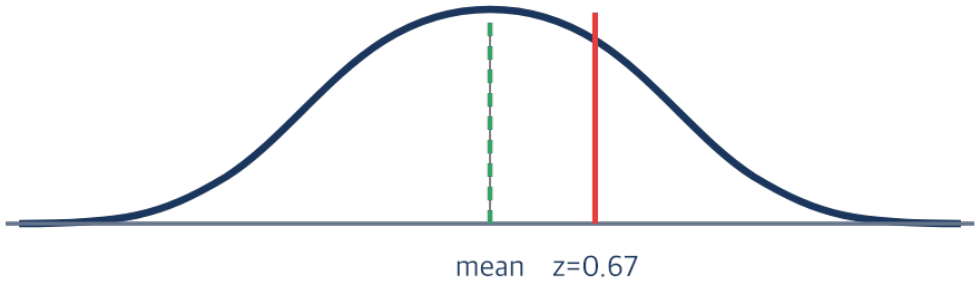
For Z ~ N(0, 1), find P(Z > 0.67).

Answer:

0.2514

Working:

Draw or imagine the right tail.



$P(Z > 0.67) = 1 - P(Z < 0.67)$.

$P(Z > 0.67) = 1 - \Phi(0.67)$.

From tables/calculator, $\Phi(0.67) \approx 0.7486$.

$1 - 0.7486 = 0.2514$.

0.2514

Teacher answers & working

Normal Distribution and Standardisation · Standard Normal

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S3

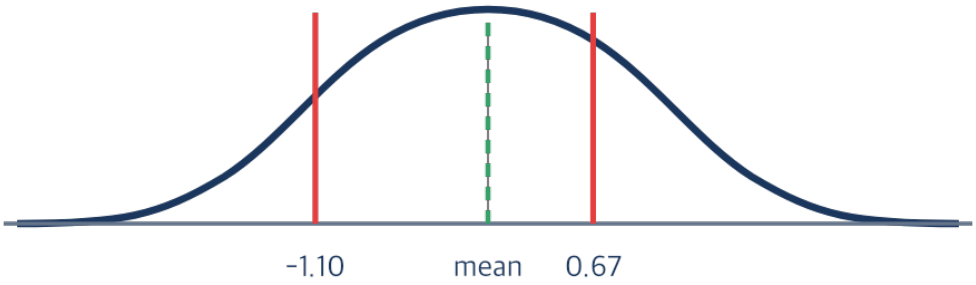
For Z ~ N(0, 1), find P(-1.10 < Z < 0.67).

Answer:

0.6129

Working:

The required region is between two z-values.



$P(-1.10 < Z < 0.67) = \Phi(0.67) - \Phi(-1.10)$.

From tables/calculator: $\Phi(0.67) \approx 0.7486$, and $\Phi(-1.10) \approx 0.1357$.

$0.7486 - 0.1357 = 0.6129$.

0.6129

Teacher answers & working

Normal Distribution and Standardisation · Standard Normal

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S4

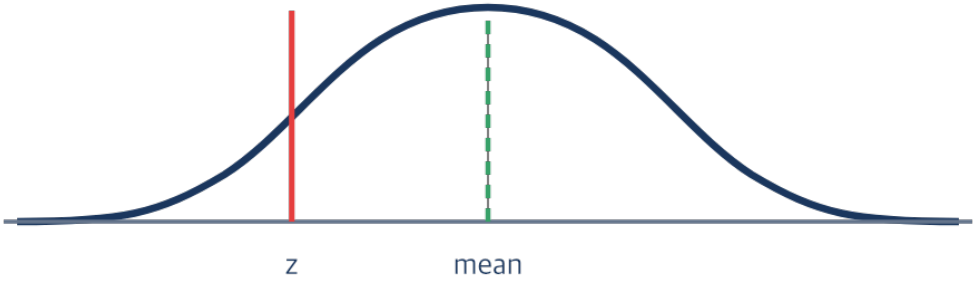
For Z ~ N(0, 1), find z such that P(Z < z) = 0.1056.

Answer:

z = -1.25

Working:

This is an inverse standard normal question.



We need Φ(z) = 0.1056.

Use inverse normal tables or a calculator: $z = \Phi^{-1}(0.1056)$.

z ≈ -1.25.

z = -1.25

Teacher answers & working

Normal Distribution and Standardisation · Standard Normal

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E1

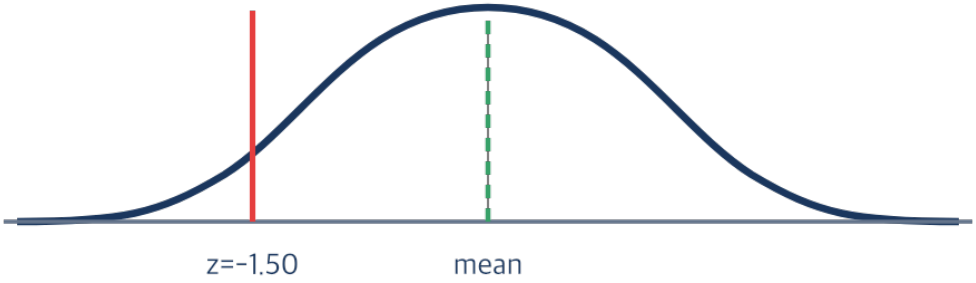
For Z ~ N(0, 1), find P(Z < -1.50).

Answer:

0.0668

Working:

Recognise this as a left-tail standard normal probability.



$P(Z < -1.50) = \Phi(-1.50)$.

Look up $\Phi(-1.50)$ using a standard normal table or calculator.

$\Phi(-1.50) \approx 0.0668$.

0.0668

Teacher answers & working

Normal Distribution and Standardisation · Standard Normal

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E2

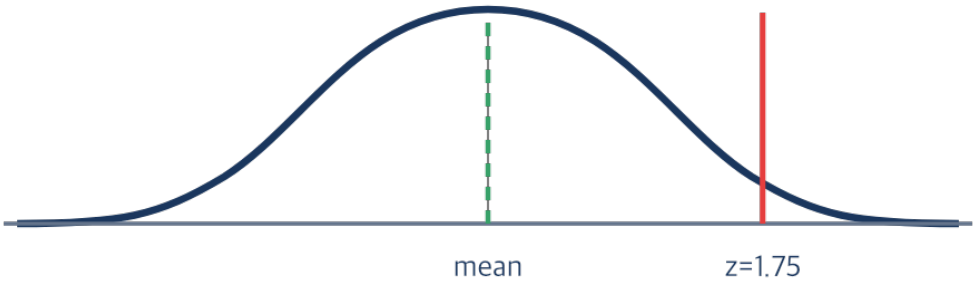
For Z ~ N(0, 1), find P(Z > 1.75).

Answer:

0.0401

Working:

Draw or imagine the right tail.



$P(Z > 1.75) = 1 - P(Z < 1.75)$.

$P(Z > 1.75) = 1 - \Phi(1.75)$.

From tables/calculator, $\Phi(1.75) \approx 0.9599$.

$1 - 0.9599 = 0.0401$.

0.0401

Teacher answers & working

Normal Distribution and Standardisation · Standard Normal

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E3

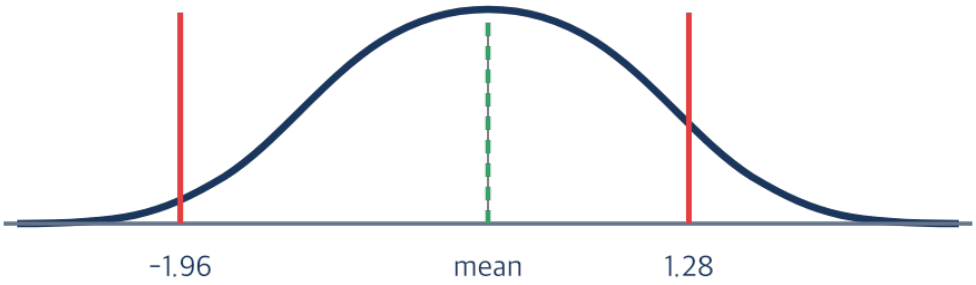
For Z ~ N(0, 1), find P(-1.96 < Z < 1.28).

Answer:

0.8747

Working:

The required region is between two z-values.



$P(-1.96 < Z < 1.28) = \Phi(1.28) - \Phi(-1.96)$.

From tables/calculator: $\Phi(1.28) \approx 0.8997$, and $\Phi(-1.96) \approx 0.0250$.

$0.8997 - 0.0250 = 0.8747$.

0.8747